

10 Basic to Intermediate Array DSA Problems

For C++ Practice

Problem 1

1. Find Second Largest Element Given an array of integers, find the second largest element. If no second largest exists, return -1.

Constraints: $1 \leq n \leq 1000$

Problem 2

2. Reverse Array In-Place Reverse the given array without using any extra space.

Constraints: $1 \leq n \leq 1000$

Problem 3

3. Find Missing Number Given array of $n-1$ numbers from 1 to n , find the missing number.

Constraints: $1 \leq n \leq 1000$

Problem 4

4. Kadane's Algorithm (Maximum Subarray Sum) Find the contiguous subarray with the largest sum.

Constraints: $1 \leq n \leq 1000$

Problem 5

5. Move All Zeros to End Move all zeros to the end of the array while maintaining relative order of non-zero elements.

Constraints: $1 \leq n \leq 1000$

Problem 6

6. Rotate Array by k Steps Rotate the array to the right by k steps in $O(n)$ time.

Constraints: $1 \leq n \leq 1000$

Problem 7

7. Find Duplicate Number Given array of $n+1$ elements with numbers from 1 to n , find the duplicate number (Floyd's Cycle Detection).

Constraints: $1 \leq n \leq 1000$

Problem 8

8. Merge Two Sorted Arrays Merge two sorted arrays into one sorted array without extra space.

Constraints: $1 \leq n \leq 1000$

Problem 9

9. Maximum Product Subarray Find the contiguous subarray with the largest product.

Constraints: $1 \leq n \leq 1000$

Problem 10

10. Trapping Rain Water Given n non-negative integers representing elevation map, compute how much water it can trap after raining.

Constraints: $1 \leq n \leq 1000$